

LOT Pager BOM

LOT® Pager — Components Buying List (v0.1)

Link legend: ✓ = opened/returned by search this session; ○ = standard vendor/part page, **not verified** — confirm part number and stock before buying. Prices are **rough estimates at 100 units, not quotes**.

A. Prototype kit (buy first, ~US\$150)

Item	Part	Link	Qty
MCU + round LCD dev board	Waveshare ESP32-S3-LCD-1.28 (GC9A01, 16 MB flash, 2 MB PSRAM)	✓ https://waveshare.com/ESP32-S3-LCD-1.28.htm	2
Touch variant w/ IMU + LiPo charger	Waveshare ESP32-S3-Touch-LCD-1.28	✓ https://www.waveshare.com/product/esp32-s3-touch-lcd-1.28.htm	2
Camera	OV2640 DVP module (24-pin FPC)	○ https://www.adafruit.com/ (search OV2640)	3
Env sensor breakout	BME688 (AI gas)	○ https://www.adafruit.com/ (search BME688)	2
Qi receiver	Adafruit Universal Qi Receiver (BQ51013B) 5 V 500 mA	✓ https://www.adafruit.com/product/1901	2
Qi transmitter pad	any 5 W Qi pad	○	2
Thin LiPo	3.7 V 100–150 mAh, ≤3 mm	○	4
Tactile switch	low-profile SMT, ≤2 mm	○	10

B. Production BOM per unit (~100 qty)

Block	Part	Est. \$	Link
MCU	ESP32-S3-WROOM-1-N16R8 (pre-certified, octal PSRAM for camera)	6	○ https://www.espressif.com/en/products/modules
Display	1.28" round IPS 240×240, GC9A01	5	✓ (via Waveshare board ref)
Camera	OV2640 2 MP module	4	○
Env sensor	Bosch BME688 (BME280 is cheaper non-gas fallback)	9	✓ BME280 https://lcsc.com/product-detail/Humidity-Sensors-Temperature-and-Humidity-Sensors_Bosch-Sensortec-BME280_C92489.html · datasheet https://cdn.sparkfun.com/assets/5/3/0/f/7/BME280_Datasheet.pdf
Light	VEML7700	1.5	○

Block	Part	Est. \$	Link
IMU	LSM6DS3TR-C or ICM-42670	2	○
Wireless power	BQ51013B + coil (see module)	4	✓ https://www.adafruit.com/product/1901
Charger / gauge	BQ25180 + MAX17048	2.5	○
Battery	thin LiPo with protection	3	○
Alert	piezo sounder + optional LRA + DRV2605L	2	○
Button	SMT tactile	0.3	○
Passives/connectors/ESD		2	
Electronics subtotal		≈ 41	
PCB 4-layer, 40×40	PCBWay	2	https://www.pcbway.com
SMT assembly (turnkey)	PCBWay, quote after upload (1–2 days)	6	✓ per search
Stainless 2-part CNC + polish	PCBWay CNC (304/316L)	20–35	✓ per search; ○ quote
Window glass/sapphire, gaskets, screws		3	
Packaging + Qi pad		6	
Estimated unit total		≈ US\$80–100	

One-off NRE (stencil, test jig, programming fixture, polishing fixture, certification budget): **~\$800–3,000 + FCC/CE if sold.**

C. Sourcing notes

- Supply MPNs; ask PCBWay for turnkey sourcing. Keep Gerbers/BOM/CPL in hardware/ when created.
- Tooling for polished stainless drives cost — get a quote for 100 vs 500 pieces.
- Check LCSC/Digi-Key stock for BME688 and ESP32-S3-WROOM before design freeze.